Assigment I pg 2 - Kelly Nobles

Question C: Discreet EV question.

- the average withdrawl amount is the long-term average, not a discreet amount that is observed w/ each transaction. EV is the mean of the withdrawl distribution which is useful for planning + forecasting cash needs in the ATM.

Question D: Prob. that amt w/drawn is w/in 1 std. dev. of the μ.

$\mu = \$50$ $\quad$ $\$50 \pm 1$ std dev.

$\sigma = \$26.27$ $\quad$ $50 - 26.27 = \underline{23.73}$

$50 + 26.27 = \underline{76.27}$

$P(23.73 \leq X \leq 76.27)$

Only 40 + 60 fall w/in $\mu \pm 1$ std. dev.

Add their probabilities.

$P(40) = 0.30$
$P(60) = \underline{0.15}$ } sum
0.45

$\boxed{45\% \text{ probability that w/drawl falls w/ } \mu \pm 1 \text{ std. de}}$

Question E: What is prob. w/drawl will fall w/in 2 std. de

$\mu = 50$ $\quad$ 50 ± 2 std dev. $\quad$ $2(26.27) = 52.54$

$\sigma = 26.27$ $\quad$ $50 - 52.54 = -2.54$

$50 + 52.54 = 102.54$

$P(-2.54 \leq X \leq 102.54)$

↳ All values fall in this range so probability is 1 or $\boxed{100\%}$

$\boxed{P(\text{w/in 2 std dev.}) = 1.00 = 100\%}$